

EventFlow AI — Project Guide

A short guide for setup, review, testing and future error fixing.

1. What this project does

- EventFlow AI is an event management app with one control center.
- It manages attendees, venues, speakers, sessions, sponsors, incidents, alerts and event insights.
- It also adds an intelligence layer with specialist agents and a central agent router.

2. Main parts

- Organizer Command Center — one place to open all five milestone cards.
- Event Intelligence Engine — shows event data, connected sources and AI functions.
- Executive Dashboard — gives a quick view with KPIs, charts, sponsor value, attendance and incident status.
- Agent Orchestration — routes tasks to Venue, Speaker, Sponsorship, Incident, Analytics and Alert agents.
- End-to-End Testing — runs route, database, agent and workflow checks.
- Production Deployment — checks release items and keeps this guide for future review.

3. Executive Dashboard

- Top health score gives a fast event view.
- KPI cards show registrations, check-ins, sponsors, open incidents, alerts and sponsor value.
- Charts show attendance, incident status, sponsor committed vs delivered value, sponsor package mix and recent registrations.
- Use the dashboard first when you need a quick event status.

4. Agent Orchestration

- The central router reads a task and picks the best agent using simple rules.
- Example: sponsor task → Sponsorship Agent; microphone issue → Incident Agent; venue task → Venue Agent.
- Run one task to see the route, or run all agents to see the full cycle.
- The routing is local and does not need an outside AI key for the demo flow.

5. Data used

- Attendees store registration, attendance, check-in time and status.
- Sponsors store package, budget, committed value, delivered value, leads, impressions, engagements and payment data.
- Incidents store title, category, severity, status, owner and time fields.
- Operational alerts store priority, source, message and read state.
- Venues, speakers and sessions support event planning.

6. How to run

- Install the packages from requirements.txt.

- Create the MySQL database using database/schema.sql.
- Check config.py and your local MySQL settings.
- Run the Flask app with Python.
- Log in as an organizer and open the Milestone Control Center.

7. Before production

- Put DB passwords and other secrets in environment variables.
- Change the Flask secret key to a strong secret.
- Use HTTPS and a production WSGI server.
- Keep database backups.
- Enable app logs and alert monitoring.
- Run the End-to-End Testing page before release.

8. If an error happens

- Page does not open — check the Flask terminal and route name.
- Database error — check MySQL is running and config.py values.
- No chart data — check that the related table has rows.
- Agent does not route — check AGENT_ROUTING_RULES in app.py.
- PDF not opening — check docs/EventFlow_AI_Project_Guide.pdf exists.
- After a code change, restart Flask and run the test page again.

9. Useful files

- app.py — main Flask app, routes, data reads and agent logic.
- config.py — local database and mail settings.
- templates/ — page files.
- static/css/style.css — page styles.
- database/schema.sql — database setup.
- README.md — project notes.
- docs/EventFlow_AI_Project_Guide.pdf — this review and error guide.

Quick review checklist

Area	Check
Login	Organizer login works
Milestones	All five cards open
Dashboard	KPIs and charts load
Agents	Route task and Run all work
Testing	Test page gives clear pass/fail
Database	MySQL tables load
Deployment	Readiness scan works
Docs	README and PDF are present

Keep this file with the project. It is meant to be a simple future reference when reviewing the work or fixing an error.